

1. Connect your laptop and a friend's phone to the same Wi-Fi network, and then use the ping command to check if both devices can communicate with each other.

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Command Prompt
Microsoft Windows [Version 10.0.19045.7417]
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C:\Users\rohan>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=50ms TTL=64
Reply from 192.168.1.1: bytes=32 time=8ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 50ms, Average = 15ms

C:\Users\rohan>
```

```
Docs:      https://doc.termux.com
Community: https://community.termux.com

Working with packages:
- Search:  pkg search <query>
- Install: pkg install <package>
- Upgrade: pkg upgrade

Report issues at https://bugs.termux.com

$ ping 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=6.52
ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=6.13
ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=6.06
ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=64 time=5.55
ms
64 bytes from 192.168.1.1: icmp_seq=5 ttl=64 time=15.5
ms
64 bytes from 192.168.1.1: icmp_seq=6 ttl=64 time=14.5
ms
64 bytes from 192.168.1.1: icmp_seq=7 ttl=64 time=13.1
ms
64 bytes from 192.168.1.1: icmp_seq=8 ttl=64 time=18.4
ms
64 bytes from 192.168.1.1: icmp_seq=9 ttl=64 time=15.2
ms
64 bytes from 192.168.1.1: icmp_seq=10 ttl=64 time=41.
3 ms
64 bytes from 192.168.1.1: icmp_seq=11 ttl=64 time=14.
6 ms
64 bytes from 192.168.1.1: icmp_seq=12 ttl=64 time=11.
4 ms
64 bytes from 192.168.1.1: icmp_seq=13 ttl=64 time=8.7
2 ms
64 bytes from 192.168.1.1: icmp_seq=14 ttl=64 time=14.
4 ms
64 bytes from 192.168.1.1: icmp_seq=15 ttl=64 time=10.
2 ms
64 bytes from 192.168.1.1: icmp_seq=16 ttl=64 time=5.3
7 ms
64 bytes from 192.168.1.1: icmp_seq=17 ttl=64 time=71.
6 ms
64 bytes from 192.168.1.1: icmp_seq=18 ttl=64 time=67.
8 ms
64 bytes from 192.168.1.1: icmp_seq=19 ttl=64 time=60.
2 ms
64 bytes from 192.168.1.1: icmp_seq=20 ttl=64 time=63.
5 ms
64 bytes from 192.168.1.1: icmp_seq=21 ttl=64 time=17.
8 ms
```

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

 - There are 30 hops it takes to reach the destination.

```
Command Prompt
Microsoft Windows [Version 10.0.19045.7417]
(c) Microsoft Corporation. All rights reserved.

C:\Users\rohan>tracert www.youtube.com

Tracing route to www.youtube.com [2404:6800:4000:101f::be]
over a maximum of 30 hops:

  0  1 ms  1 ms  1 ms  undefined.hostname.localhost [2402:a00:400:815:6231:92ff:fe16:8c45]
  1  54 ms  2 ms  4 ms  undefined.hostname.localhost [2402:a00:404:1::1111]
  2  3 ms  6 ms  5 ms  undefined.hostname.localhost [2402:a00:1::3d]
  3  13 ms  12 ms  12 ms  undefined.hostname.localhost [2402:a00::a]
  4  58 ms  13 ms  13 ms  2404:6800:8281:340::1
  5  13 ms  13 ms  12 ms  2404:6800:8281:340::1
  6  13 ms  13 ms  13 ms  2801:4860:0:1::2130
  7  13 ms  14 ms  13 ms  2801:4860:0:1::16e20
  8  * * * Request timed out.
  9  * * * Request timed out.
 10  * * * Request timed out.
 11  * * * Request timed out.
 12  * * * Request timed out.
 13  * * * Request timed out.
 14  * * * Request timed out.
 15  14 ms  14 ms  13 ms  lcbomo-in-f190.1e100.net [2404:6800:4000:101f::be]

Trace complete.

C:\Users\rohan>
```

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

 - It displays error message of “General failure”

```
Command Prompt
Microsoft Windows [Version 10.0.19045.7417]
(c) Microsoft Corporation. All rights reserved.

C:\Users\rohan>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
General failure.
General failure.
General failure.
General failure.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\rohan>
```

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

IP Address	MAC address	Likely Device	Person/Owner
192.168.1.1	60-31-92-16-8c-45	Router/Default Gateway	Shared
192.168.31.254	00-50-56-e0-ac-6f	Virtual Network Adapter	Unknown
192.168.231.254	00-50-56-fd-b1-c7	Virtual Network Adapter	Unknown

5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Network Setup and Connectivity Testing Report

Introduction

- The purpose of this task was to set up a basic computer network, verify communication between devices, and analyse the network's performance using the ping and traceroute commands. These tools help confirm connectivity and identify the path that data takes across the network.

Step-1 Network Setup

- I began by connecting the network devices according to the required topology. The devices included computers connected through a switch, with a router used to provide communication between different networks. Appropriate Ethernet cables were used to establish the physical connections.
- Next, I configured the IP settings for each device. Each computer was assigned a unique IP address within the same subnet, along with the correct subnet mask and default gateway. After completing the configuration, I verified that all devices had the correct network settings and that the interfaces were active.

Step-2 Testing Connectivity Using Ping

- Once the network was configured, I tested connectivity using the ping command. First, I pinged the local loopback address (127.0.0.1) to confirm that the TCP/IP stack was functioning correctly. Then, I pinged the IP address of another computer on the network.
- The ping results showed successful replies with minimal response times and no packet loss. This confirmed that the devices could communicate successfully across the network. The command also displayed statistics such as packets sent, packets received, packet loss percentage, and average round-trip time.

Step-3 Testing the Route Using Traceroute

- After verifying basic connectivity, I used the tracert command to identify the path taken by packets to reach the destination device. The command displayed each network hop between the source and destination, along with the response time for each hop.
- In this network, the tracert output showed only a small number of hops because the devices were connected within a simple local network. Each hop responded successfully, indicating that routing was functioning as expected.

Observations and Issues

- The network setup was completed successfully, and all configured devices were able to communicate with one another. The ping command returned consistent responses with no packet loss, indicating a stable connection. The tracert command confirmed that packets followed the expected path through the network.
- During the setup process, one issue occurred when a device was initially assigned an incorrect IP address, preventing communication. After correcting the IP configuration and ensuring that all devices were in the correct subnet, connectivity was restored. No further issues were observed.